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Mon, Aug 17 at 7:22 AM

You the expert in CFA level of Knowledge , I will provide you some question related to Derivatives topic , i want you to answer with your best level of knowledge.

Absolutely. I'll answer your **CFA Level I Derivatives** questions at an exam-focused level.

For each question, I can give you:

1. **Correct answer** — A/B/C
2. **Why it is correct** — from the CFA Level I perspective
3. **Why the other choices are wrong**
4. **Key concept/rule** to remember
5. **Formula**, whenever applicable
6. **Exam shortcut / trap** — what CFA is testing
7. **Simple numerical example**, when it helps

I'll also distinguish carefully between concepts such as:

- Forward vs. futures
- Long vs. short positions
- Payoff vs. profit
- Value vs. price
- Calls vs. puts
- Hedging vs. speculation vs. arbitrage
- Options vs. forwards/futures
- Put–call parity
- No-arbitrage pricing
- Forward pricing
- Futures pricing
- Swaps
- Interest-rate derivatives
- Equity, currency, commodity, and bond derivatives

Send me your **first Derivatives question**.

Montau AG is a German capital goods producer that manufactures its products
domestically and delivers its products to clients globally. Montau's global sales
manager shares the following draft commercial contract with his Treasury team:
Montau AG Commercial Export Contract
Contract Date: [Today]
Goods Seller: Montau AG, Frankfurt, Germany
Goods Buyer: Jeon Inc., Seoul, Korea
Description of Goods: A-Series Laser Cutting Machine

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Correct Answer: B

Montau AG has entered into a **75-day forward commercial contract**:

- Today → Contract is signed.
- 75 days later → Montau delivers the machine and receives **KRW 650 million**.
- Montau's costs are primarily in **EUR**.
- Therefore, Montau will be exposed to changes in the **KRW/EUR exchange rate over the 75-day period**.

The key reason to use a **derivative rather than a spot transaction** is the **timing mismatch**: Montau does not receive the KRW until 75 days from today.

Let's understand the risk

Montau will receive:

$$KRW\,650,000,000$$

in **75 days**.

But Montau ultimately needs **EUR**.

Suppose today:

$$1\ EUR = 1,600\ KRW$$

Then the expected EUR value is:

$$\frac{650,000,000}{1,600} = EUR406,250$$

But Montau doesn't receive the KRW today.

If, after 75 days:

$$1\ EUR = 1,800\ KRW$$

then:

$$\frac{650,000,000}{1,800} = EUR361,111$$

Montau receives **EUR 45,139 less** than it would have received at today's exchange rate.

So Montau has **foreign exchange risk**.

Why a derivative?

Montau could enter into a **forward FX contract today** to sell KRW and buy EUR in 75 days.

Conceptually:

Today

$$\text{Enter FX forward}$$

⬇️

75 days

$$\text{Receive KRW 650 million}$$

⬇️

$$\text{Deliver KRW under forward}$$

⬇️

$$\text{Receive predetermined EUR}$$

This locks in the EUR value of the future KRW receipt.

Why B is correct

> **B. Montau AG faces a 75-day timing difference between the commercial contract date and the delivery date when Montau AG is paid for the machine in KRW.**

This is exactly what creates the need for a derivative.

A **spot FX transaction** settles essentially immediately, but Montau doesn't have the KRW today. It will receive the KRW **75 days later**.

Therefore, an **FX forward** is appropriate.

Why A is not the best answer

> **A. Montau AG is selling a machine at a contract price in KRW and incurs costs based in EUR.**

This identifies the **source of the currency exposure**, but it doesn't explain why a **derivative rather than a spot transaction** is needed.

The crucial issue is that the KRW payment occurs **75 days in the future**.

Why C is incorrect

> **C. Montau AG is unable to sell KRW today...**

This is not the fundamental issue.

Montau doesn't have KRW today because the buyer hasn't paid yet. The appropriate solution is to **contract today for a future FX exchange**, rather than needing to sell KRW immediately.

⭐ CFA Level I Exam Rule

> **Future foreign-currency receipt/payment → consider an FX forward.**

Think:

Future cash flow + FX uncertainty = Derivative hedge

Whereas:

Cash available today + immediate FX conversion = Spot transaction

Quick memory trick

SPOT = NOW

FORWARD = LATER

Here:

$$\boxed{\text{KRW received in 75 days}}$$

Therefore:

$$\boxed{\text{FX Forward}}$$

Answer: B ✅

2. Which of the following types of derivative and underlyings are best suited to
hedge Montau's financial risk under the commercial transaction?
A. Montau AG should consider a firm commitment derivative with currency as
an underlying, specifically the sale of KRW at a fixed EUR price.
B. Montau AG should consider a contingent claim derivative with the price of
the machine as its underlying, specifically an A-series

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Correct Answer: A — Firm Commitment Derivative with Currency as the Underlying ✅

This question is testing **two things**:

1. **What type of derivative?** → Firm commitment or contingent claim?
2. **What is the underlying?** → Currency or machine?

Step 1: Identify Montau's exposure

Montau will receive:

$$\boxed{KRW\ 650,000,000 \text{ in } 75 \text{ days}}$$

But Montau is a German company, so it ultimately needs **EUR**.

Therefore, Montau is exposed to **KRW/EUR exchange-rate risk**.

The appropriate hedge is to **sell KRW and receive EUR at a predetermined exchange rate in 75 days**.

That is an **FX forward**.

Step 2: Is a forward a firm commitment or contingent claim?

This is the key CFA distinction.

Firm commitment derivative

A firm commitment derivative creates an **obligation** for both parties.

Examples:

- Forward
- Futures
- Swap

Contingent claim

A contingent claim gives the **holder a right, but not an obligation**.

Main example:

- Option

Montau knows with certainty that it expects to receive KRW 650 million under the commercial contract.

It therefore wants to **lock in** the exchange rate.

So a **forward** is appropriate.

$$\boxed{\text{Forward} = \text{Firm commitment}}$$

Step 3: What is the underlying?

The financial risk is **currency risk**, not the price of the laser machine.

Therefore:

$$\boxed{\text{Underlying} = \text{Currency}}$$

Montau should arrange a contract to:

$$\boxed{\text{Sell KRW and receive EUR}}$$

at a predetermined exchange rate.

Evaluate the choices

A. Correct ✅

> Firm commitment derivative with currency as the underlying, specifically the sale of KRW at a fixed EUR price.

This describes an **FX forward**.

Montau:

- will receive KRW in the future;
- wants to eliminate exchange-rate uncertainty;
- can enter into a forward today;
- will sell the future KRW proceeds for EUR.

Therefore:

$$\boxed{\text{A is correct}}$$

B. Incorrect ❌

> Contingent claim derivative with the price of the machine as its underlying.

The machine's price is **not the financial risk Montau needs to hedge**.

The contract already specifies:

$$\boxed{KRW\ 650,000,000}$$

The uncertainty is what that KRW amount will be worth in **EUR**.

Also, a contingent claim means an **option**, which is not the most direct hedge here.

C. Incorrect ❌

> Contingent claim derivative with currency as the underlying, specifically the sale of EUR at a fixed KRW price.

There are **two problems**:

1. Wrong derivative type

A contingent claim is an option. Montau has a predictable future KRW receipt and wants to lock in the exchange rate, so a **firm commitment/forward** is more appropriate.

2. Wrong currency direction

Montau will receive **KRW**, so it needs to hedge by:

$$\boxed{\text{Selling KRW / Buying EUR}}$$

not selling EUR.

⭐ CFA Level I Decision Rule

When you see a question like this, use:

What is the exposure?

Future foreign currency receipt/payment

⬇️

Does the company want to lock in the exchange rate?

Yes

⬇️

Firm commitment

$$\boxed{\text{Forward/Futures}}$$

⬇️

What is the underlying?

$$\boxed{\text{Currency}}$$

⬇️

Direction?

Future **KRW receipt**

$$\boxed{\text{Sell KRW / Buy EUR}}$$

Final answer

$$\boxed{\textbf{A}}$$

Memory rule:

> **Future foreign-currency cash flow + desire to lock in exchange rate = FX forward (firm commitment).**

If Montau enters into a centrally cleared derivative contract on the OTC market, which of the following statements about credit risk associated with the derivative is most likely correct? A. Montau faces credit risk associated with the possibility that its counterparty to the contract may not fulfill its contractual obligation. B. Montau poses a credit risk to its counterparty because it may fail to fulfill its contractual obligation. C. Montau poses a credit risk to a derivative contract end user holding a contract with the opposite features of Montau's.

Correct Answer: B ✅

This question is testing an important distinction in **centrally cleared OTC derivatives**: **who is exposed to whom for credit risk?**

Step 1: What happens when an OTC derivative is centrally cleared?

Normally, in an OTC derivative:

$$\text{Montau} \longleftrightarrow \text{Counterparty}$$

Each party faces the possibility that the other party defaults.

But with **central clearing**, a **central counterparty (CCP)** stands between the two original parties.

Conceptually:

$$\text{Montau} \longleftrightarrow \boxed{\text{CCP}} \longleftrightarrow \text{Original Counterparty}$$

The CCP becomes the buyer to every seller and the seller to every buyer.

Therefore, Montau's **direct counterparty credit risk is to the CCP**, rather than to the original OTC counterparty.

Step 2: Why is B correct?

> **B. Montau poses a credit risk to its counterparty because it may fail to fulfill its contractual obligation.**

This is the best answer.

Under central clearing, Montau's contractual relationship is effectively with the **central counterparty**.

If Montau fails to meet its obligations—for example, fails to provide required payments or collateral—**Montau represents credit risk to the CCP**.

So:

$$\boxed{\text{Montau's default} \rightarrow \text{credit risk to CCP}}$$

Therefore, **B is correct.**

Why A is wrong ❌

> Montau faces credit risk associated with the possibility that its counterparty may not fulfill its contractual obligation.

This would generally describe a **bilateral OTC derivative that is not centrally cleared**.

For an uncleared OTC derivative:

$$\text{Montau} \leftrightarrow \text{Counterparty}$$

Montau faces counterparty credit risk.

But with central clearing:

$$\text{Montau} \leftrightarrow \text{CCP} \leftrightarrow \text{Counterparty}$$

The CCP substantially changes the counterparty-credit-risk structure.

Why C is wrong ❌

> Montau poses a credit risk to a derivative contract end user holding a contract with the opposite features of Montau's.

This misunderstands the role of the **CCP**.

Montau does **not directly face another end user** holding the opposite position.

Instead:

$$\boxed{\text{CCP stands between the two parties}}$$

So Montau's failure creates credit risk for the **CCP**, not directly for the other end user.

⭐ CFA Level I Exam Rule

OTC derivative

Without central clearing:

$$A \longleftrightarrow B$$

Both parties face **counterparty credit risk**.

Centrally cleared OTC derivative:

$$A \longleftrightarrow CCP \longleftrightarrow B$$

The CCP becomes the counterparty to both sides.

Therefore:

> **Each original party poses credit risk to the CCP, and the CCP manages the resulting counterparty risk.**

Quick memory trick

No CCP → I worry about YOU.

CCP → CCP worries about BOTH of us.

$$\boxed{\text{Answer} = \text{B}}$$

I did not get this .

Absolutely. This is a **tricky CFA question**, so let's forget the complicated terminology and build it from **zero**.

1. First, forget "OTC" and "centrally cleared"

Imagine two people make a deal:

Montau and **Bank A**

Montau says:

> "I will pay/receive according to our derivative contract in the future."

Bank A says:

> "Okay."

Now ask:

What happens if Bank A doesn't pay Montau?

Montau loses money.

So:

$$\boxed{\text{Bank A can create credit risk for Montau}}$$

This is what **Answer A** is describing.

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2. Now introduce a CCP

With **central clearing**, there is a middleman called a:

Central Counterparty (CCP)

Instead of:

```
Montau  ←──────────→  Bank A
```

we have:

```
Montau  ←─────→  CCP  ←─────→  Bank A
```

The CCP stands **in the middle**.

This is the most important thing to understand.

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3. What does the CCP actually do?

Suppose Montau has a contract with Bank A.

The CCP effectively says:

> "You don't need to worry directly about each other. I will stand between you."

So:

```
Montau
   ↓
   CCP
   ↓
Bank A
```

The CCP becomes:

- **buyer to every seller**
- **seller to every buyer**

---

4. Now ask: Who is Montau's counterparty?

This is the key to the question.

Without central clearing:

$$\boxed{\text{Montau's counterparty} = \text{Bank A}}$$

With central clearing:

$$\boxed{\text{Montau's counterparty} = \text{CCP}}$$

Therefore, Montau's direct counterparty credit risk is now related to the **CCP**, not directly to Bank A.

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5. Now look at the question again

> **If Montau enters into a centrally cleared derivative contract...**

That word **"centrally cleared"** is the giveaway.

It means:

```
Montau ⟷ CCP ⟷ Other party
```

Now consider what happens if **Montau itself defaults**.

Montau doesn't fulfill its obligation.

Who is affected?

The CCP.

Therefore:

$$\boxed{\text{Montau can create credit risk for the CCP}}$$

And that's exactly what **B** says:

> **B. Montau poses a credit risk to its counterparty because it may fail to fulfill its contractual obligation.**

Here, **"its counterparty" = CCP**.

Therefore:

$$\boxed{\text{B is correct}}$$

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6. Why isn't A correct?

A says:

> Montau faces credit risk because **its counterparty may not fulfill its obligation.**

The problem is that in a centrally cleared derivative, the original counterparty is no longer directly facing Montau.

Instead:

```
Montau ⟷ CCP ⟷ Original Counterparty
```

So the question is testing whether you understand the **CCP's role**.

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7. A very simple real-life analogy

Imagine you and your friend make a bet.

Without a middleman

You:

> "I'll pay you ₹1,000 if I lose."

Your friend:

> "Okay."

If your friend wins and **you don't pay**, your friend has a problem.

If your friend loses and **doesn't pay**, you have a problem.

You directly face each other.

With a middleman

Now imagine a trusted person says:

> "Don't pay each other directly. Give everything to me. I'll make sure the winner gets paid."

Now:

```
You ⟷ Trusted person ⟷ Friend
```

If **you don't fulfill your obligation**, the **trusted person** has the problem.

If **your friend doesn't fulfill their obligation**, the **trusted person** has the problem.

That's basically the CCP concept.

8. The CFA exam shortcut ⭐

Whenever you see:

"Centrally cleared derivative"

Immediately think:

$$\boxed{\text{CCP stands between the two parties}}$$

Then:

If Montau defaults:

$$\boxed{\text{Montau creates credit risk for CCP}}$$

If the other party defaults:

$$\boxed{\text{Other party creates credit risk for CCP}}$$

So:

```
          CCP
         /   \
        /     \
   Montau    Other Party
```

Both parties have a credit relationship with the **CCP**.

One sentence to remember for the exam

> **In a centrally cleared derivative, the CCP becomes the counterparty to both sides, so each party's default creates credit risk for the CCP.**

That's why:

$$\boxed{\mathbf{B}}$$

If you want, I can also explain **why central clearing reduces counterparty credit risk but does NOT eliminate credit risk**—that's another CFA concept that often appears immediately after this one.

ChatGPT can make mistakes. Check important info.